Emma Oliver

I'm a double data science and music major at Oregon State University's Honors College interested in public opinion and survey research. I have experience in data visualization and survey methodology, and I have two publications using data analysis in wood science, one in press and one currently in review.

Education

DATA SCIENCE & MUSIC (DOUBLE MAJOR) **2024 – Present**
Honors College | Oregon State University

- 4.0 GPA
- Advanced Data Science option
- Expected graduation June 2028

Experience

SURVEY RESEARCH CENTER TECH ASSISTANT **June – September 2026**
Department of Statistics and Data Science | Oregon State University

- Produced survey visualizations in R to communicate survey findings clearly and accurately to non-technical audiences
- Assisted in survey research work for two major mixed mode surveys by proof-reading and editing survey reports and creating detailed comparisons of surveys across years

UNDERGRADUATE RESEARCHER **January – September 2026**
Department of Wood Science and Engineering | Oregon State University

- Authored an interdisciplinary research paper on curve fitting of biochar Raman spectroscopy data
- Collaborated with an international research group on a publication on wood curve comparison

REEU STUDENT RESEARCHER **June – September 2025**
Department of Wood Science and Engineering | Oregon State University

- Conducted literature review, identified research gaps, and performed research on biochar Raman spectroscopy curve fitting sensitivity to variable models, bounds, and starting parameters
- Communicated research findings in a 20-minute presentation to a combined technical and non-technical audience

COLLABORATIVE PIANIST **2021 – Present**
Corvallis, OR

- Perform both solo and collaboratively as an advanced classical pianist, including a past member of the Corvallis-OSU Symphony Orchestra
- Coordinate OSU Piano Department's social media account to bring attention to the program
- Volunteered for 100+ hours as a community pianist and worked as a piano accompanist for both a solo vocalist and a theater camp

Publications

Oliver, Emma, Samuel Olajide, Silvana Calvano, Francesco Negro, and Islam Hafez. "Raman spectroscopy curve fitting methods for biochar and related carbonaceous materials." (In review)

Calvano, Silvana, Francesco Negro, Emma Oliver, Islam Hafez, and Alessandro Portaluri. "Curve Comparison in Wood Science: An Introductory Guide to Common Cases." 2026. *Wood and Fiber Science*. (Accepted for publication)

Awards and Honors

- Featured "Meet Data Science Major" for OSU College of Science's social media channels **2026**
- Junior Waldo-Cummings Outstanding Student Award **2025-2026**
- Drucilla Shepard Smith Scholastic Award **2024-2025 | 2025-2026**

Skills

- R
- Python
- Data visualization
- Communication with technical and non-technical audiences
- Survey methods
- Data analysis
- Teamwork, including international collaboration